LD for the Quarter = (4*2% + 2*4% + 1*6%)/90 = 22/90 = 0.24%

38.5 SLAs

38.5.1 SLA 001

Metric Type	Definition	Target	LD Applicable
Accuracy (De-duplication) Measurement interval (Daily)	*False Positive Identification Rate (FPIR):* A term applying to de-duplication transactions only. It is the ratio of the number of false positive identification decisions to the total number of de-duplication transactions for which no duplicate exists in the gallery, regardless of database size.	<= 0.5%	None.
		> 0.5 %	Biometric solution kept in observation till next quarter and if the FPIR doesn't get lesser than 0.5%, may impact allocation of de-duplication transaction volumes till the Biometric solution becomes compliant to the SLA.
	False Negative Identification Rate (FNIR): A term applying to de-duplication transactions only. It is the ratio of the number of false negative identification decisions to the total number of de-duplication transactions for which a duplicate exists in the gallery, regardless of database size. **FNIR @ FPIR = 0.5%**	<=0.1%	None
		>0.1% and <= 0.2%	2% of applicable quarterly payment May impact allocation of de-duplication transaction volumes till the Biometric solution becomes compliant to the SLA
		> 0.2% and <= 0.5%	4% of applicable quarterly payment May impact allocation of de-duplication transaction volumes till the Biometric solution becomes compliant to the SLA

		> 0.5%	6% of applicable quarterly payment Biometric Solution may be kept under Probation till the Biometric solution becomes compliant to the SLA. All de-duplication transactions processed by the BSP while the BSP is kept under probation shall be non-payable and shall be used for accuracy measurement only.

Measurement methodology

FPIR is the ratio of the number of false positive identification decisions to the total number of de-duplication transactions for which no duplicate exists in the gallery.

FNIR is the ratio of the number of false negative identification decisions to the total number of de-duplication transactions for which a duplicate exists in the gallery.

SLA Calculation:

False Negative Identification Rate (FNIR) = (Number of False Negative Identification decisions) / (Number of de-duplication transactions for which a duplicate exists in the gallery)

False Positive Identification Rate (FPIR) = Number of False Positive Identification decisions / (Number of de-duplication transactions for which no duplicate exists in the gallery)

While the Accuracy (FNIR and FPIR) shall be measured daily, the calculation of LD shall be done on quarterly basis using following formula

[((Number of days FNIR was >0.1% and <=0.2%) X 2%) + ((Number of days FNIR was >0.2% and <=0.5%) X 4%) + ((Number of days FNIR was >0.5%) X 6%)] / Number of Days in the Quarter

38.5.2 SLA 002

Deleted

38.5.3 SLA 003

Metric Type	Definition	Target	LD applicable
De-duplication throughput Measurement interval (daily)	*De-duplication throughput:* ABIS should be able to parallelize de-duplication requests and shall be able to de-duplicate at-least 350,000 de-duplications per day.	>= 350,000	None
		<350,000 but >= 300,000	1% of applicable quarterly payment
		<300,000 but >= 250,000	2% of applicable quarterly payment
		<250,000 but >= 200,000	3% of applicable quarterly payment
		< 200,000	5% of applicable quarterly payment Biometric Solution may be kept under Probation till their throughput becomes more than 200,000 de-duplications / day. All de-duplication transactions processed by the BSP while the BSP is kept under probation shall be non-payable and shall be used for accuracy measurement only.

Measurement methodology

This SLA is aimed at finding de-duplication throughput (DT) per day. The ABIS shall be able to process a minimum of 350,000 de-duplications requests per day. **The number of deduplication requests will be the number of requests sent to the queue for processing that have been in the queue for more than 24 hours**

Note: If the number of de-duplications given to ABIS is less than 350,000 for any given day, than for that day this SLA shall not apply if the bidder is able to process more than 98% of the requests on the same day, otherwise an LD of 1% shall be applicable for that day.

SLA Calculation:

While the throughput of de-duplication shall be measured daily, the calculation of LD shall be done

on quarterly basis using following formula: [((Number of days when Deduplication requests were < 350,000 and processed by ABISs were less than 98%) * 1%) + (Number of days when DT was < 350,000 and >=300,000) * 1%) + ((Number of days when DT was < 300,000 and >=250,000) * 2%) + ((Number of days when DT was < 250,000 and >=200,000) * 3%) + ((Number of days when DT was < 200,000) * 5%)]/ Number of Days in the Quarter

38.5.4 SLA 004

Metric Type	Definition	Target	LD applicable
Accuracy Measurement interval (daily)	*False Positive Identification Rate for Anomalous matches (FPIRA):* A term applying to de-duplication transactions only. It is the ratio of the number of false positive anomalous identification decisions to the total number of de-duplication transactions for which no match exists in the gallery, regardless of database size	<= 5%	None.
		> 5 %	Biometric solution kept in observation till next quarter and if the FPIRA doesn't get lesser than 5%, may impact allocation of transaction volumes till the Biometric solution becomes compliant to the SLA.
	False Negative Identification Rate for Anomalous matches (FNIRA): A term applying to de-duplication transactions only. It is the ratio of missed anomalous identification decisions to the total number of de-duplication transactions for which anomalous match exists in the enrolment database, regardless of database size **FNIRA @ FPIRA = 5%**	<= 1%	None
		> 1% and <= 2%	1% of applicable quarterly payment May impact allocation of de-duplication transaction volumes till the Biometric solution becomes compliant to the SLA

		> 2% and <= 5%	2% of applicable quarterly payment May impact allocation of de-duplication transaction volumes till the Biometric solution becomes compliant to the SLA
		> 5%	3% of applicable quarterly payment Biometric Solution may be kept under Probation till the Biometric solution becomes compliant to the SLA. All de-duplication transactions processed by the BSP while the BSP is kept under probation shall be non-payable and shall be used for accuracy measurement only.

Measurement methodology

Anomalous false positives are the false positive anomalous identification decisions when no match exists in the gallery.

Anomalous false negatives are the false negatives where no instance of any modality was found matching but it had one or more instance of a modality in common across the gallery (but not a True Duplicate).

SLA Calculation:

False Positive Identification Rate-Anomalous (FPIRA) = Number of false positive anomalous identification decisions / (Number of de-duplication transactions for which no match exists in the gallery)

False Negative Identification Rate-Anomalous (FNIRA) = Number of missed anomalous identification decisions / (Number of de-duplication transactions for which anomalous match exists in the gallery)

While the Accuracy (FNIRA and FPIRA) shall be measured daily, the calculation of LD shall be done on quarterly basis using following formula

[((Number of days FNIRA was >1% and <=2%) X 1%) + ((Number of days FNIRA was >2% and <=5%) X 2%) + ((Number of days FNIRA was >5%) X 3%)] / Number of Days in the Quarter

38.5.5 SLA 005

Metric Type	Definition	Target	LD applicable
Uptime Measurement interval (5 minutes)	Uptime of ABIS Solution	>=99%	None
		>=98% and <99%	3% of applicable quarterly payment
		>=96% and <98%	4% of applicable quarterly payment

		< 96%	5% of applicable quarterly payment Biometric Solution may be kept under Probation till their uptime becomes more than 99%. All de-duplication transactions processed by the BSP while the BSP is kept under probation shall be non-payable and shall be used for accuracy and uptime measurement only

Measurement methodology

Process to capture raw data for SLA calculations: The ABIS solution will be considered as "Up" if during every 5 minute interval the ABIS had a request (like de-duplication or insert) to process in any of the queue and it processed at least one such request in each of those queues. ABIS shall be considered "Up" for that time period when MSIP Planned/unplanned downtime notification is available

SLA Calculation:
Uptime for a day = {1- [(Total Downtime) / (Total Time - Planned Downtime)]} *100

where
Total Downtime is the number of 5 minute intervals in the day when the ABIS was not "Up"
Total Time is the number of 5 minute intervals in the day (12*24=288)
Planned Downtime is the number of 5 minute intervals in the day when the ABIS had a planned downtime

While the uptime shall be measured daily, the calculation of LD shall be done on quarterly basis using following formula
[((Number of days uptime was >=98% and <99%) X 3%) + ((Number of days uptime was >=96% and <98%) X 4%) + ((Number of days uptime was <96%) X 5%)] / Number of days in the Quarter

Note: For computation of uptime SLAs of ABIS, downtime due to data center environment planned and unplanned downtimes (i.e., power and HVAC downtime) and planned and unplanned downtimes of Compute, Storage and Network infrastructure when the BSP solution is deployed on As-Is infrastructure and IaaS and PaaS services when the BSP solution is deployed on To-Be infrastructure shall be excluded.

38.5.6 SLA 006

Metric Type	Definition	Target	LD applicable
Accuracy for Fingerprint Measurement interval (Daily)	*Bona Fide Presentation Classification Error Rate (BPCER):* Proportion of bona fide presentations incorrectly classified as presentation attacks. A term applying	<= 50%	None.

	to quality assessment transactions only. It is the ratio of the number of false positives while finding the attempts to defraud during biometric acquisition to the total number of biometric acquisitions transactions where no attempt has been made to defraud.	> 50 %	Biometric solution kept in observation till next quarter and if the BPCER doesn't get lesser than 50%, will impact allocation of transaction volumes till the Biometric solution becomes compliant to the SLA.
	Attack Presentation Classification Error Rate (APCER): Proportion of missed presentation attacks. A term applying to quality transactions only. It is the ratio of the number of false negatives finding the attempts to defraud during biometric acquisition to the total number of biometric acquisitions transactions where an attempt has been made to defraud **APCER@ BPCER= 50%**	<=1%	None
		>1% and <=5%	2% of applicable quarterly payment May impact allocation of de-duplication transaction volumes till the Biometric solution becomes compliant to the SLA
		>5%	4% of applicable quarterly payment Biometric Solution may be kept under Probation till the Biometric solution becomes compliant to the SLA. All de-duplication transactions processed by the BSP while the BSP is kept under probation shall be non-payable and shall be used for accuracy measurement only.

Measurement methodology

This SLA is aimed at finding attempts to defraud during biometric acquisition and shall include usage of Non-human / synthetic biometrics, usage of intermediate or proximal phalanx of fingers instead of distal phalanx, vertical placement of fingers on the sensor causing very little portion of the tip of the finger to get capture, etc.

SLA Calculation:

Attack Presentation Classification Error Rate (APCER) = (Number of misses finding the attempts to defraud during biometric acquisition using a presentation attack / Number of biometric acquisitions transactions where an attempt has been made to defraud using a presentation attack) *100

Bona Fide Presentation Classification Error Rate (BPCER) = (Number of false positives finding the attempts to defraud during biometric acquisition through a presentation attack / Number of biometric acquisitions transactions with no attempts to defraud) * 100

While the BPCER and APCER shall be measured daily, the calculation of LD shall be done on quarterly basis using following formula
[((Number of days APCER was >1% and <=5%) X 2%) + ((Number of days APCER was >5%) X 4%)] / Number of Days in the Quarter

38.5.7 SLA 007

Metric Type	Definition	Target	LD applicable
Minimum Resource Availability Measurement Interval (Monthly)	This SLA applies to the availability of all resources in a month. The attendance logged by each of these resources will be used to compute the monthly availability.	Monthly availability for each of the project resources, including applicable leaves and UIDAI holidays, is defined below;	
		>= 16 business days AND >=144 hours per month	None
		>=12 and <16 business days AND >=108 and <144 hours per month	2% of applicable quarterly payment
		< 12 business days and < 108 hours	4% of applicable quarterly payment

Measurement methodology

Calculation of actual business days and total number of hours logged (on UIDAI biometric attendance system) per month for all resources.

38.5.8 SLA 008

Metric Type	Definition	Target	LD applicable
Minimum overlap period during resource replacement Measurement Interval (Quarterly)	This SLA is to measure the “overlap period” between an outgoing BSP resource and the incoming BSP resource. A minimum of 2 business weeks is necessary to ensure adequate knowledge transition and handover of project tasks from the outgoing resource to the incoming resource.	Overlap period >=2 business weeks	0% of applicable quarterly payment for every ‘1’ replacement
		Overlap period >= 1 business week and < 2 business weeks	1% of applicable quarterly payment for every ‘1’ replacement
		Overlap period < 1 business week	2% of applicable quarterly payment for every ‘1’ replacement

<u>Measurement methodology</u>

Overlap Period = Number of business weeks where the outgoing and incoming resource(s) are present in UIDAI premises, as per the UIDAI biometric attendance system.

Where

Business Week = Monday to Friday, in a week (including applicable holidays as per UIDAI holiday calendar)

38.5.9 SLA 009

Metric Type	Definition	Target	LD applicable
Number of Resource replacements in a Quarter Measurement Interval (Quarterly)	Number of resource replacements for “Resources” as per Clause 8 in Volume-II of RFP	‘1’ replacement	None
		More than 1 replacement	1% of applicable quarterly payment for every ‘1’ replacement

<u>Measurement methodology</u>

The SLA will be calculated based on the number of resource replacement exercised by the BSP in the quarter.

38.5.10 SLA 010

Metric Type	Definition	Target	LD applicable
Accuracy for Fingerprint Measurement interval (when the SLA targets are revised)	*False Match Rate (FMR):* A term applying to authentication transactions only. The ratio of number of false matches that are resulted by matching probes (authentication data) with non-mated references in the gallery (enrolment images) to the total number of such matches attempted.	<=0.01%	None
		>0.01%	5% of applicable quarterly payment
	False Non- Match Rate (FNMR): A term applying to authentication transactions only. The ratio of number of false non- matches that are resulted by matching probes (authentication data) with mated references in the gallery (enrolment images) to the total number of such matches attempted. FNMR will be computed after the SDK meets the requirement of FMR FNMR @ FMR = 0.01%	<=2%	None
		>2% and <= 3%	0.2% of applicable quarterly payment
		>3%	0.4% of applicable quarterly payment

Metric Type	Definition	Target	LD applicable

Measurement methodology

Process to capture raw data for SLA calculations: Ground truth verified Data collected by UIDAI will be used as probes for calculating the accuracy of the SDKs.

SLA Calculation: Expectation is for the SDK to be able to match the probes collected during data capture with their enrolment records. If SDK is not able to match a probe with its mated enrolment pair, then it would constitute a false non- match. Similarly, if the probes are matched with non-mated enrolment records and result in a match, it would constitute a false match.

False Match Rate (FMR) = (Number of False Matches) / (Number of probes when match is attempted with non-mated references)
False Non- Match Rate (FNMR) = (Number of False non-matches) / (Number of probes when match is attempted with mated references)

The ground truth verified data shall be collected by UIDAI.

This exercise of finding if the BSP has submitted a SDK that meets the FNMR <2 % @ FMR = 0.01% will be done before any of the SDKs is chosen to be used in the production. Any SDK not matching the above accuracy criterion (FNMR < 2% @ FMR = 0.01%) shall be considered non-compliant and shall attract the necessary LD as per this SLA. Once the SDKs are found to be compliant to the above criterion, this exercise shall not be repeated. However, UIDAI shall keep reviewing the improvements in the state-of-the-art algorithms once a year based on their submissions in the NIST and other such reputed organizations who publish the accuracy of all such participating algorithms. If the UIDAI finds that the state-of-the-art has improved, the SLA criteria might be suitably changed and each of the participating BSP shall be given one quarter to submit new SDKs compliant to the new SLAs. If any of the BSP shall not submit a SDK that is compliant to the new SLAs within a quarter, it shall be considered non-compliant to this SLA and shall attract the necessary LD each quarter as per this SLA.

Note: This SLA is applicable only for fingerprint authentication

38.5.11 SLA 011

Metric Type	Definition	Target	LD applicable
Backup & Restore – Scheduled Backups Measurement interval (Monthly)	Compliance to the backup schedule as agreed upon by the UIDAI	100%	None
		<100%	2% of applicable quarterly payment

Measurement methodology

Assumption:

i. SLA will be applicable when the BSP solution's UAT / Go-Live has been provided.
ii. There may be cases wherein the SLA data is not captured in the respective automated / reporting tool due to unforeseen reasons. In such cases, appropriate artefacts / evidences will be considered for SLA calculation.
iii. There may be scenarios where BSP may not be responsible for the breach of the SLA. These scenarios may include the dependencies beyond the control of BSP, and / or managed by UIDAI or other ecosystem partners. For e.g., unavailability of power, services not managed by BSP, planned downtime, renewal of licenses by UIDAI, etc. Such cases will be exempted from SLA calculations provided appropriate artefacts / evidences are provided by BSP.
iv. Any hardware components, applications, portals, and services which are configured in redundant / HA mode at either one or across multiple DCs, the downtime will be considered when all the redundant components, application, portals and services are down simultaneously, and the services get impacted
v. BSP to ensure the timely availability of data for scheduled incremental and full backups on a daily, weekly, monthly basis as per UIDAI's backup policy.

Data Points for SLA Calculation:

a. Backup software logs provide relevant information of initiation and completion of backups.

38.5.12 SLA 012

Metric Type	Definition	Target	LD applicable
Monthly testing and Validation of Backup Devices Measurement Interval (Monthly)	Monthly testing and validation of backup devices for the quality of data	100%	None
		<100%	2% of applicable quarterly payment

Measurement methodology
Assumption: a. The sample size for restoration shall be taken as per backup testing and restoration process b. Restoration shall be done in the restoration environment. c. Certain fixed size of data will be restored from each backup and then given to Application owner for testing d. Backup team shall ensure that the backed-up data has been restored without any errors e. Data validation and integrity check of the restore is performed by the application owner and the confirmation to be provided to backup team. **f.** BSP to ensure the testing and validation of backup devices in collaboration with MSIP as defined in Volume II – Scope of Work. **Data Points for SLA Calculation:** • The restoration testing will be performed on sample basis as per the backup testing and restoration process. Any deviation in testing and validation of the backup devices from the testing and restoration process will result in the breach of the SLA. **Reports to be submitted:** • Sample size for the restoration testing • Restoration testing report and confirmation from respective UIDAI's stakeholders for successful restoration testing • Artefacts, if any

38.5.13 SLA 013

ABIS application and MM-SDK -related issues will be categorized as per the 'Defect Matrix' [refer the Table below], which will be used to calculate SLAs and the corresponding LD pertaining to application-related issues. The salient points of this 'Defect matrix' include the following:

1. All defects will be classified under 4 types (Blocker, Critical, Major, Minor) based on "severity".
2. Classification of a defect will be based on specific scenarios outlined in the table below. The issue tracking tool will auto-classify a defect based on these scenarios.
3. For 'Blocker', 'Critical' and 'Major' defect types, a pre-defined number of workaround scenarios will have to be adopted by the BSP as the first-course of action. The BSP is expected to ensure that these workaround scenarios are applied at the earliest, as per the turnaround time (TAT) defined in the table below. Subsequently, the BSP is also expected to fix the corresponding defect as per the specified TAT.

For all defects, the following conditions shall apply:

1. Severity of 'each' defect will be assigned by the UIDAI Helpdesk [based on end user request or type of issue logged] and can be changed only at the discretion of UIDAI.
2. All application related defects that are caused by OEM technology components, will be the sole responsibility of the BSP and shall be required to comply with the TAT, for the corresponding severity, as mentioned below.

Type	Applicable Defect Scenarios	Workaround Scenarios	Turnaround time (TAT) for Workaround	Turnaround time (TAT) for Defect Fix
Blocker	a. Application hangs/ unresponsive/ crashes; b. Application has memory leak causing restart within 24 hours; c. Application throws exception/ returns error response on more than 1% of daily requests; d. Application latency more than “5x of benchmarked value” on more than 1% of daily requests; e. Application has security issue other than zero-day defect	a. Rollback b. Restart using scripts c. Additional instances to handle load	8 hours from reporting of the defect	For defects with ‘Low’ or ‘Moderate complexity = 7 days; For defects with ‘High’ complexity = 15 days; **Note** • Defect scenarios (a) to (e) will be applicable after a ‘grace period`, for all new enhancements, at the discretion of UIDAI; • All ‘blocker’ defects will be also categorized as ‘Low’ or ‘Moderate” complexity by default. If any defect fix has dependencies on multiple modules, it may be categorized as ‘High’ complexity only with UIDAI approval
Critical	a. Application has memory leak causing restart within 7 days; b. Application throws exception/ returns error response on more than 0.1% of daily requests; c. Application latency more than 5x benchmarked value on more	i. Rollback ii. Restart using scripts iii. Additional instances to handle load	24 hours from reporting of the defect	For defects with ‘Low’ or ‘Moderate ‘complexity = 15 days; For defects with ‘High’ complexity = 30 days; **Note** • Defect scenarios (a) to (c) will be applicable after a ‘grace period`, for all new enhancements, as determined by UIDAI; • All ‘critical’ defects will be categorized

Type	Applicable Defect Scenarios	Workaround Scenarios	Turnaround time (TAT) for Workaround	Turnaround time (TAT) for Defect Fix
	than 0.1% of daily requests;			as 'Low' or 'Moderate 'complexity by default. If any defect fix has dependencies on multiple modules, it may be categorized as 'High' complexity only with UIDAI approval;
Major	a. Application throws exception/ returns error response on less than 0.1% of daily requests; b. Application latency more than 5x benchmarked value on less than 0.1% of daily requests;	i. Rollback ii. Restart using scripts iii. Additional instances to handle load	24 hours from reporting of the defect	For defects with 'Low' or 'Moderate' complexity = 30 days; For defects with 'High' complexity = 45 days; **Note** • Defect scenarios (a) to (b) will be applicable after a 'grace period`, for all new enhancements, as determined by UIDAI; • All 'major' defects will be categorized as 'Low' or 'Moderate 'complexity by default. If any defect fix has dependencies on multiple modules, it may be categorized as 'High' complexity only with UIDAI approval

Type	Applicable Defect Scenarios	Workaround Scenarios	Turnaround time (TAT) for Workaround	Turnaround time (TAT) for Defect Fix
Minor	a. Low-impact issues (Incorrect UI labels, Incorrect validation messages, etc.) in the applications. b. All Information requests will also be classified under this category.	Not applicable	Not applicable	For defects with ‘Low’ or ‘Moderate’ complexity = 45 days; For defects with ‘High’ complexity = 60 days; **Note** All ‘minor; defects will be categorized as ‘Low’ or ‘Moderate ‘complexity by default. If any defect fix has dependencies on multiple modules, it may be categorized as ‘High’ complexity only with UIDAI approval;

Metric Type	Definition	Defect Severity	Target	LD applicable
TAT for defects Measurement Interval (Monthly)	Turn Around Time (TAT) for the fixing of defects, identified in the Production environment and classified as per the ‘Defect matrix’ above	Blocker	<= 7 days (for complexity = ‘Low’ or ‘Moderate’) <=15 days (for complexity = ‘High’)	0
			Within additional 5 working days	2% of applicable quarterly payment
		Critical	<= 15 days (for complexity = ‘Low’ or ‘Moderate’) <= 30 days (for complexity = ‘High’)	0
			Within additional 5 working days	1% of applicable quarterly payment
		Major	<= 30 days (for complexity = ‘Low’ or ‘Moderate’) <= 45 days (for complexity = ‘High’)	0

			Within additional 5 working days	0.5% of applicable quarterly payment

Measurement methodology

Data Points for SLA Calculation:

1. Variance = (Actual time in days taken to resolve the defect from the date of reporting) – (Expected TAT for defect, based on defect severity and complexity level).
2. The result will be rounded off, for determining the variance.
3. Any delay beyond additional 5 working days: An additional 0.1% of LD will be applicable for every day of delay beyond the specified timeline unless a specific waiver has been provided on the milestone by UIDAI for factors causing delay which are outside the BSP's control.

Reports to be submitted:

Automated [using the issue reporting date and resolution date, captured from the 'Issue tracking' tool]

38.5.14 SLA 014

Milestone based SLAs

- The table below shows the various activities and timelines which are to be treated as key project milestones and the SLA Levels corresponding to them.
- The BSP shall use Project Management Tool to continuously track and monitor the project progress along with dependencies.
- The BSP shall submit a weekly status report along with the updated report in the Project Management Tool.
- The SLAs linked to milestones are applied once on slippage of the specified target timeline.
- For milestone based SLAs, in addition to the one-time application of applicable LD \with the milestone, an additional 0.1% of LD will be applicable for every day of delay beyond the specified timeline unless a specific waiver has been provided on the milestone by UIDAI for factors causing delay which are outside the BSP's control.
- For Milestone 1 and 2, The LD will be applicable on the first payment that BSP will receive from UIDAI.
- For Milestone 3, 4 and 5, the LD will be applicable on the invoice post completion and acceptance of the milestone.

S. No.	Category	Description	Target for completion (T being Signing of contract or Notification from UIDAI for Project Initiation whichever is later)	Reporting tools to be used	Reports/data to be submitted	LD applicable
1	Templatization	Templatization of all the existing records in the	T+ 3 months	Project Management Tool	Completion Report	2% of applicable

S. No.	Category	Description	Target for completion (T being Signing of contract or Notification from UIDAI for Project Initiation whichever is later)	Reporting tools to be used	Reports/data to be submitted	LD applicable
		enrolment and update database				quarterly payment
2	Commissioning and Go-Live	Go-live	T+ 6 months	Project Management Tool	UAT Test Plan and Acceptance Testing Report	2% of applicable quarterly payment
3	Migration to the To-Be infrastructure	Migration of BSP solution to the To-Be private cloud infrastructure of UIDAI from the As-Is infrastructure of UIDAI	3 months from the date of notification of readiness of the To-Be infrastructure from UIDAI	Project Management Tool	UAT Test Plan and Acceptance Testing Report	2% of applicable quarterly payment
4	MM-SDK	Android and iOS support in MM-SDK	T+24 months	Project Management Tool	UAT Test plan and acceptance testing report of MM-SDK for Android and iOS	2% of applicable quarterly payment
5	GPU compatibility	ABIS solution should be compatible with GPU	T+24 months	Project Management Tool	UAT Test plan and acceptance testing report of GPU Compatibility	2% of applicable quarterly payment

38.5.15 SLA 015

Metric Type	Definition	Target	LD applicable
Accuracy for Iris Measurement interval (when the SLA targets are revised)	*False Match Rate (FMR):* A term applying to authentication transactions only. The ratio of number of false matches that are resulted by matching probes (authentication data) with non-mated references in the gallery (enrolment images) to the total number of such matches attempted.	<=0.001%	None
		>0.001%	5% of applicable quarterly payment
	False Non- Match Rate (FNMR): A term applying to authentication transactions only. The ratio of number of false non- matches that are resulted by matching probes (authentication data) with mated references in the gallery (enrolment images) to the total number of such matches attempted. FNMR will be computed after the SDK meets the requirement of FMR FNMR @ FMR = 0.001%	<=1%	None
		>1% and <= 2%	0.2% of applicable quarterly payment
		>2%	0.4% of applicable quarterly payment

Metric Type	Definition	Target	LD applicable

Measurement methodology

Process to capture raw data for SLA calculations: Ground truth verified Data collected by UIDAI will be used as probes for calculating the accuracy of the SDKs.

SLA Calculation: Expectation is for the SDK to be able to match the probes collected during data capture with their enrolment records. If SDK is not able to match a probe with its mated enrolment pair, then it would constitute a false non- match. Similarly, if the probes are matched with non-mated enrolment records and result in a match, it would constitute a false match.

False Match Rate (FMR) = (Number of False Matches) / (Number of probes when match is attempted with non-mated references)

False Non- Match Rate (FNMR) = (Number of False non-matches) / (Number of probes when match is attempted with mated references)

The ground truth verified data shall be collected by UIDAI.

This exercise of finding if the BSP has submitted a SDK that meets the FNMR <1 % @ FMR = 0.001% will be done before any of the SDKs is chosen to be used in the production. Any SDK not matching the above accuracy criterion (FNMR < 1% @ FMR = 0.001%) shall be considered non-compliant and shall attract the necessary LD as per this SLA. Once the SDKs are found to be compliant to the above criterion, this exercise shall not be repeated. However, UIDAI shall keep reviewing the improvements in the state-of-the-art algorithms once a year based on their submissions in the NIST and other such reputed organizations who publish the accuracy of all such participating algorithms. If the UIDAI finds that the state-of-the-art has improved, the SLA criteria might be suitably changed and each of the participating BSP shall be given one quarter to submit new SDKs compliant to the new SLAs. If any of the BSP shall not submit a SDK that is compliant to the new SLAs within a quarter, it shall be considered non-compliant to this SLA and shall attract the necessary LD each quarter as per this SLA.

Note: This SLA is applicable only for Iris authentication

38.5.16 SLA 016

Metric Type	Definition	Target	LD applicable
Accuracy for Face Measurement interval (when the SLA targets are revised)	*False Match Rate (FMR):* A term applying to authentication transactions only. The ratio of number of false matches that are resulted by matching probes (authentication data) with non-mated references in the gallery (enrolment images) to the total number of such matches attempted.	<=0.001%	None
		>0.001%	5% of applicable quarterly payment

Metric Type	Definition	Target	LD applicable
	False Non- Match Rate (FNMR): A term applying to authentication transactions only. The ratio of number of false non- matches that are resulted by matching probes (authentication data) with mated references in the gallery (enrolment images) to the total number of such matches attempted. FNMR will be computed after the SDK meets the requirement of FMR FNMR @ FMR = 0.001%	<=1%	None
		>1% and <= 2%	0.2% of applicable quarterly payment
		>2%	0.4% of applicable quarterly payment

Measurement methodology

Process to capture raw data for SLA calculations: Ground truth verified Data collected by UIDAI will be used as probes for calculating the accuracy of the SDKs.

SLA Calculation: Expectation is for the SDK to be able to match the probes collected during data capture with their enrolment records. If SDK is not able to match a probe with its mated enrolment pair, then it would constitute a false non- match. Similarly, if the probes are matched with non-mated enrolment records and result in a match, it would constitute a false match.

False Match Rate (FMR) = (Number of False Matches) / (Number of probes when match is attempted with non-mated references)

False Non- Match Rate (FNMR) = (Number of False non-matches) / (Number of probes when match is attempted with mated references)

The ground truth verified data shall be collected by UIDAI.

This exercise of finding if the BSP has submitted a SDK that meets the FNMR <1 % @ FMR = 0.001% will be done before any of the SDKs is chosen to be used in the production. Any SDK not matching the above accuracy criterion (FNMR < 1% @ FMR = 0.001%) shall be considered non-compliant and shall attract the necessary LD as per this SLA. Once the SDKs are found to be compliant to the above criterion, this exercise shall not be repeated. However, UIDAI shall keep reviewing the improvements in the state-of-the-art algorithms once a year based on their submissions in the NIST and other such reputed organizations who publish the accuracy of all such participating algorithms. If the UIDAI finds that the state-of-the-art has improved, the SLA criteria might be suitably changed and each of the participating BSP shall be given one quarter to submit new SDKs compliant to the new SLAs. If any of the BSP shall not submit a SDK that is compliant to the new SLAs within a quarter, it shall be considered non-compliant to this SLA and shall attract the necessary LD each quarter as per this SLA.

Note: This SLA is applicable only for Face authentication